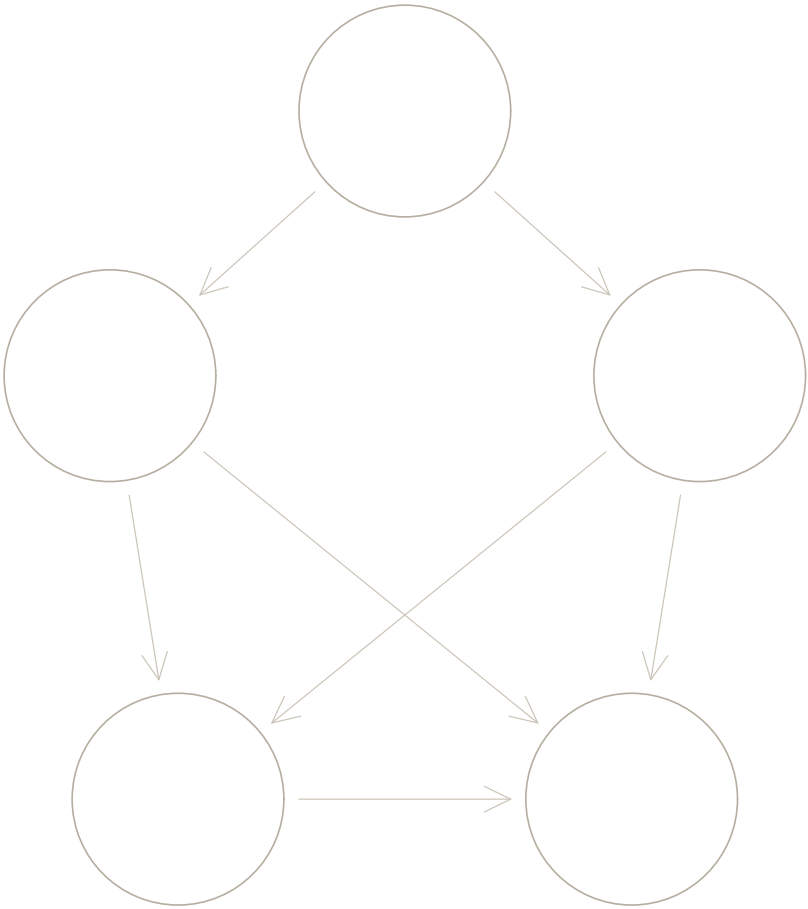Cross-linked cause network